VAMSI KRISHNA BH

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SUMMARY

Senior Software Engineer with 6+ years of experience designing and delivering scalable, cloud-native, event-driven systems for enterprise and telecom platforms. Expert in Java 17, Spring Boot, Microservices, and WebFlux, with a deep foundation in distributed systems, fault tolerance, and system design. Led large-scale modernization initiatives transforming legacy monoliths into resilient microservices using Kafka, RabbitMQ, and PostgreSQL, driving major gains in scalability and release velocity. Full-stack engineer with strong AWS, Docker, and Kubernetes expertise enabling automated CI/CD pipelines and zero-downtime deployments.

TECHNICAL SKILLS

- **Backend:** Java (11–17), Spring Boot, Microservices, WebFlux, REST APIs, Hibernate, JPA, Distributed Systems
- **Frontend:** React.js, Angular (12–16), Redux Toolkit, TypeScript, JavaScript, HTML5, CSS3
- **Cloud & DevOps:** AWS (EC2, S3, RDS, ECR, ECS, Lambda, API Gateway, IAM), Docker, Kubernetes, Jenkins, CI/CD, Unleash
- **Messaging:** Apache Kafka, RabbitMQ, IBM MQ, WebSockets, Event-Driven Architecture
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, ORM, Database Design
- **Testing:** JUnit, Mockito, Jasmine, REST Assured, TDD
- **Tools**: Git, Maven, Gradle, JIRA, Confluence, Swagger/OpenAPI, ELK Stack

PROFESSIONAL EXPERIENCE

Senior Software Engineer — Verizon | TX, USA | Feb 2025 – Present

- Led modernization of large-scale telecom platforms, transforming legacy monoliths into event-driven microservices using Java 17, Spring Boot, WebFlux, Kafka, and AWS — significantly improving scalability and system resilience.
- Designed and delivered 30+ REST APIs across distributed services, reducing code duplication by 40% and accelerating feature development and release cycles.
- Implemented asynchronous messaging workflows using Kafka, RabbitMQ, and IBM MQ, enabling seamless interoperability between modern microservices and legacy systems during the TracFone integration.
- Built high-performance, real-time UIs using React.js and Redux Toolkit, reducing UI latency by ~40% and measurably improving end-user experience.
- Optimized PostgreSQL and MySQL queries, indexing strategies, and data access patterns, improving reporting and retrieval performance by 30%.
- Improved application reliability and security by implementing resilience patterns, JWT-based authentication, RBAC, and centralized logging with the ELK Stack, consistently supporting 99.99% uptime.

Software Engineer — SLHT Services | NJ, USA | Jan 2023 – Feb 2025

- Built and supported enterprise workforce management applications across the full SDLC using Java, Spring Boot, Spring Security, and microservices architecture.
- Developed secure backend services with REST APIs, JWT authentication, and role-based authorization, improving system security and compliance posture.
- Built data-driven dashboards and admin interfaces using Angular (12–16), React.js, Highcharts, and Chart.js, enabling real-time reporting and operational insights.
- Designed and optimized PostgreSQL/MySQL schemas, SQL/PL-SQL queries, and indexing strategies, significantly improving reporting performance and data accuracy.
- Deployed and maintained microservices on AWS (EC2, ECS, RDS, S3, Lambda, API Gateway) using Docker, Kubernetes, and Jenkins, improving deployment reliability and release turnaround time.

Graduate Research & Teaching Assistant — Columbus State University | Columbus, GA | Jan 2022 – Dec 2022

- Designed and implemented energy-efficient, hardware-secured communication protocols for IoT devices, focusing on mutual authentication and edge security.
- Presented research at ACM Conference — architected a hardware-secured mutual authentication protocol using Zymkey to establish trusted device identity across distributed IoT and edge-computing systems.
- Served as Teaching Assistant for Python, Java, SQL, and Database Design; led Robotics Programming and Python with Cybersecurity summer camps for students.
- Recognized with the Academic Excellence Award for research contributions and instructional impact.

Software Developer — Silicon Matrix | Hyderabad, India | Jun 2019 – Aug 2021

- Developed and enhanced frontend features for a high-traffic e-commerce platform using React, Redux, JavaScript, HTML5, and CSS3.
- Implemented and consumed REST APIs using Java, Spring Boot, and JWT authentication, supporting secure CRUD operations and core business workflows.
- Identified and resolved SQL bottlenecks, improving API response times and overall backend reliability.
- Built reusable UI components with React Hooks and contributed to modernization of legacy Spring MVC applications to Spring Boot and cloud deployments.

EDUCATION

M.S. in Applied Computer Science
Columbus State University, Georgia · Graduate Research & Teaching Assistant

B.Tech in Computer Science
Malla Reddy College of Engineering and Technology, India

CERTIFICATIONS & ACHIEVEMENTS

AWS Certified Solutions Architect – Associate
Academic Excellence Award — Columbus State University
Selected as **Graduate Research & Teaching Assistant based on academic performance and technical expertise**

PUBLICATIONS & RESEARCH

- Master's Thesis: Towards Energy-Efficient Edge Computing for Tiny AI Applications
  - Researched and implemented edge-optimized AI inference workflows, focusing on efficient deployment, orchestration, and secure communication for distributed AI-enabled systems.
- **ACM Conference: Mutual Authentication Protocol between IoT Devices and Hardware Security using ZYMKEY — Gatlinburg, TN (2022)**
  - Architected a hardware-secured mutual authentication protocol using Zymkey to establish trusted device identity and secure communication across distributed IoT and edge-computing systems.